

Exponents and Scientific Notation36 points · two periods; students continue the same paper **ANSWER****KEY***Show your work. Circle your final answer.***This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.***If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.***1. Laws of Exponents**

9 points

1. Write each expression as a single power.

a. $4^3 \cdot 4^5$ 1 pt
 4^8

Add the exponents; the base stays 4.

b. $(2^3)^4$ 1 pt
 2^{12}

Multiply the exponents.

c. $\frac{7^9}{7^4}$ 1 pt
 7^5

Subtract the exponents.

d. 9^0 (give its value) 1 pt
 1

*Any nonzero base to the zero power is 1.**Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.*

2. Find the value of each expression.

a. $\left(\frac{2}{3}\right)^3$ 1 pt

$\frac{8}{27}$
b. $6^0 \cdot 5^2$ 1 pt

25

$1 \cdot 25$. A 0 here means 6^0 was read as 0.

c. $\frac{(10^2)^2}{10^3}$ 1 pt

10

$10^4 \div 10^3 = 10^1$.

Scoring: a 1 · b 1 · c 1 = 3 points.

3. Which expression is equivalent to $2^4 \cdot 2^3$? 1 pt

☐ A. 4^7

☐ B. 2^{12}

☐ C. 4^{12}

☒ D. 2^7

D

Scoring: 1 point.

4. Find the value of n that makes the statement true: $\frac{5^n}{5^2} = 125$ 1 pt

$n = 5$

$125 = 5^3$, so $n - 2 = 3$.

Scoring: 1 point.

2. Integer Exponents

9 points

5. Rewrite each expression with positive exponents. Do not evaluate.

a. 6^{-3} 1 pt

$$\frac{1}{6^3}$$

b. $\frac{1}{(-2)^{-5}}$ 1 pt

$$(-2)^5$$

The power moves up; the base keeps its sign and its parentheses.

c. $3 \cdot 10^{-4}$ 1 pt

$$\frac{3}{10^4}$$

The 3 is a factor, not a base with an exponent; it stays up.

Scoring: a 1 · b 1 · c 1 = 3 points.

6. Find the value of each expression.

a. 4^{-2} 1 pt

$$\frac{1}{16}$$

Reciprocal of 4^2 . -16 is the misconception; -8 multiplied base by exponent.

b. $(-3)^{-3}$ 1 pt

$$-\frac{1}{27}$$

1 over $(-3)^3 = 1$ over -27.

c. $\left(\frac{1}{2}\right)^{-3}$ 1 pt

$$8$$

Flip to 2, then cube.

d. $2^5 \cdot 2^{-7}$ 1 pt

$$2^{-2} = \frac{1}{4}$$

$$5 + (-7) = -2.$$

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

7. Which is the value of 5^{-2} ? 1 pt

☐ A. $\frac{1}{10}$

☐ B. -25

☒ C. $\frac{1}{25}$

☐ D. -10

C

Scoring: 1 point.

8. Select ALL of the expressions whose value is $\frac{1}{64}$. 1 pt

- ☒ A. 2^{-6}
☒ B. 4^{-3}
☐ C. -8^2
☐ D. 2^{-8}
☒ E. $\left(\frac{1}{4}\right)^3$

A, B and E

All three correct and nothing else for the point.

Scoring: 1 point.

3. Variable Bases

8 points

9. Write each expression as a single monomial.

a. $x^4 \cdot x^6$ 1 pt

x^{10}

b. $(y^3)^5$ 1 pt

y^{15}

c. $(3a^2)^3$ 1 pt

$27a^6$

The 3 is cubed as well. $3a^6$ and $9a^6$ are the two coefficient errors.

d. $\frac{15m^7}{5m^2}$ 1 pt

$3m^5$

Coefficients divide; exponents subtract.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

10. Rewrite each expression using only positive exponents.

a. b^{-4} 1 pt

$\frac{1}{b^4}$

b. $5x^{-2}$ 1 pt

$\frac{5}{x^2}$

Only the x moves; $1/(5x^2)$ moved the coefficient too.

c. $(2m^{-3}n)^2$ 1 pt

$\frac{4n^2}{m^6}$

$4m^6n^2$, then m^6 moves down.

Scoring: a 1 · b 1 · c 1 = 3 points.

11. Which expression is equivalent to $(4x^3)^2$? 1 pt
- ☐ A. $16x^5$
- ☐ B. $8x^6$
- ☐ C. $4x^6$
- ☒ D. $16x^6$

D

Scoring: 1 point.

4. Scientific Notation

10 points

12. Write each number in scientific notation.

a. 8,300,000 1 pt

8.3×10^6

b. 0.00047 1 pt

4.7×10^{-4}

The 4 is in the fourth decimal place; 10^{-3} counted the zeros.

c. 60,500,000,000 1 pt

6.05×10^{10}

The interior zero stays in the coefficient.

d. 0.0000000092 1 pt

9.2×10^{-9}

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

13. Write each number in standard form.

a. 2.9×10^5 1 pt

$290,000$

b. 7.1×10^{-6} 1 pt

0.0000071

Scoring: a 1 · b 1 = 2 points.

14. Which is 0.0036 written in scientific notation? 1 pt

- ☐ A. 3.6×10^3
- ☒ B. 3.6×10^{-3}
- ☐ C. 3.6×10^{-2}
- ☐ D. 36×10^{-4}

B

Scoring: 1 point.

15. Answer each question.

- a. How many times larger is 9×10^9 than 3×10^6 ? 1 pt

$3 \times 10^3 = 3,000$ times

$9 \div 3 = 3$ and $10^9 \div 10^6 = 10^3$. '3 times' ignored the powers; '30' read 10^3 as 10.

- b. How many times larger is 4.8×10^{-3} than 1.6×10^{-7} ? 1 pt

$3 \times 10^4 = 30,000$ times

$4.8 \div 1.6 = 3$ and $-3 - (-7) = 4$.

Scoring: a 1 · b 1 = 2 points.

16. A computer displays a number as 5.6E8. Write the number in standard form. 1 pt

560,000,000

E8 means $\times 10^8$.

Scoring: 1 point.

Score Tracker

36 points, one point per lettered part. Section 1 — 9 · Section 2 — 9 · Section 3 — 8 · Section 4 — 10

Benchmark	Questions	Possible	Earned
MA.7.NSO.1.1	1, 2, 3, 4	9	
MA.8.NSO.1.3	5, 6, 7, 8	9	
MA.8.AR.1.1	9, 10, 11	8	
MA.8.NSO.1.4	12, 13, 14, 15, 16	10	
Total		36	

Follow-through (ruling 19). A later part is correct when the right operation is applied to the student's own earlier value. Look at what they wrote for the earlier part, do this part's operation to it, and tick if that is what they wrote. It applies only where the work is on the page.

Transfer items (ruling 18): none on this paper. Both were cut: (1) Croix, 7 October 2026 — the old question 21 (between which two consecutive powers of 10 a number lies) needed a step the unit does not teach. (2) Croix, 7 October 2026 — the old question 10 (which is greater, 2^{-3} or 3^{-2} : show how you know) went with every question answered in a sentence; he does not want to read explanations on this paper.